P1419 Effective Mechanisms for Engaging ECRs in Higher Education Policy Training

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Abstract

COVID-19 has caused a lot of misinformation with regards to the value & importance of science in dealing with the pandemic. In order to address this, connecting scientists, policymakers and lay audiences on science-related issues is essential. In order to bridge the gap between researchers and policymakers, the *Journal of Science Policy & Governance (JSPG)* in collaboration with the Graduate Professional Success in STEM (GPS-STEM) Public Policy Prep Program at UC Irvine and other partners, offered the new virtual Science Policy & Advocacy for STEM Scientists Certificate Program. The program equips early career researchers (ECRs) with fundamental skills in science policy and advocacy, including effective communication for engaging policymakers on a multitude of issues. Certificate Program learning sessions described effective communication mechanisms and methods for ECRs to engage policymakers. The program also provided ECRs with opportunities for networking, as well as peer-to-peer review and feedback from policy professionals.

Background & Program Goals

Background: Scientists must play a significant role in the post-COVID-19 recovery process by educating policymakers & the public on the importance of evidence-based research, in order to ensure the development of comprehensive, effective & long-lasting policies. This can be encouraged by providing STEM scientists with a better understanding of science policy and advocacy through specialized training. As we plan to recover from the pandemic, we also need to develop effective and ethical policies to prepare ourselves in tackling its ripple effects.

Goals: To accomplish this, University of California, Irvine's Public Policy Prep (P3) program (previously supported by Burroughs Wellcome Fund) designed the 'Science Policy & Advocacy Certificate Program for STEM Scientists' offered virtually for PhD students & postdocs.

Focus and format: The program focused on teaching basic science policy and advocacy skills and concepts, identifying concrete ways to transition into careers in these areas, and engaged students in networking opportunities. The program incorporated lectures on various topics in science policy & advocacy, workshops for developing communication skills, & peer-to-peer learning opportunities to sharpen skills in these fields.

Participants: 120 PhD students & Postdocs from multiple STEM disciplines from 8 higher education institutes registered for the course. Accounting for the attrition rate due to long-term time commitment, schedule changes, ~70 participants were eligible for certificate of course completion.

Certificate requirements: Podcast summary; elevator pitch video; STEM education survey; policy writing piece; one pager talking points; informational interview; power mapping plan; public engagement plan.

Organizers & Coordinators

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Harinder Singh, PhD, Program Director, GPS-STEM, UC Irvine
Melissa Varga, Science Network Community & Partnerships Manager, Union of Concerned Scientists (UCS)
Coordinators: Ria Deshpande & Klebea Carvalho, GPS-STEM Council Members ; **Melania Abrahamanian, PhD,** Postdoctoral Scholar, UCLA

Program:
https://gps.bio.uci.edu/science-policy-and-advocacy-for-stem-scientists/
Twitter: @SciPolUCIrvine @SciPolJournal

Curriculum

Policy Topics & Areas:

Module 1: Introduction to Science Policy & Advocacy
Module 2: Scientific Research Policy
Module 3: STEM Education & Workforce Development

Advocacy & Policy Fellowships:

Module 4: Effective Advocacy Strategies for Policymakers
Module 5: AAAS STPF for Transitioning into Science Policy & Advocacy Careers
Module 6: State and Federal Policy Fellowships for Policy & Advocacy

Policy Skills & Industry Careers:

Module 7: Effective Communication for Public Policy & Advocacy
Module 8: Leveraging Science Policy & Advocacy for Non-academic and Industry Careers

Local Engagement & Advocacy in Action:

Module 9: Science Policy & Advocacy Engagement at the Local Level
Module 10: Science Policy & Advocacy in Action

Policy Topics & Areas

These modules explained fundamentals of science policy and advocacy, explored approaches for researching and pitching on topics in these areas, and exposed students to policy topics on scientific research topics and STEM education & workforce development policy focused on improving the research enterprise by increasing equity and inclusion.

Module 2: Scientific Research Policy. Speakers: Kei Koizumi (AAAS, JSPG), Steve Allison (Ridge2Reef), Sara Marriott (George Mason U.), Chanel Matney (CA State Assembly), Adriana Bankston (JSPG), and Laura Lasiter (Friends of Cancer Research). Moderator: Ben Wolfson (NCI, JSPG).

Advocacy & Policy Fellowships

These modules discussed what advocacy means, covered ways of interacting with policymakers at the federal and state level, and discussed several options for policy fellowships through which STEM scientists can transition into science policy and advocacy careers.

Module 5: AAAS STPF for Transitioning into Science Policy & Advocacy Careers. Speakers: Cynthia Bernardez (AAAS STPF), Ellen Wann (NIH), Marguerite Matthews (NINDS), Teresa M. Stoepler (InterAcademy Partnership), Stephanie Bailey (Chapman University), Benjamin Isaacoff (U.S. Department of State, JSPG). Moderator: Leah Cairns (NASEM).

Program Activities & Outcomes

Activities:

- **Listen** to podcasts to learn about science policy careers
- **Construct** science policy elevator pitch for policymakers
- **Write** op-ed or policy memo on topic of interest
- **Design** advocacy one pager document for policymakers
- **Craft** a power mapping plan with targeted messages for policymakers

Outcomes:

- **Build** expertise in giving a science policy pitch to policymakers
- **Grow** your network among peers and other science policy experts
- **Perform** informational interviews with science policy professionals
- **Learn** fundamentals of science policy and advocacy from various resources

Policy Skills & Industry Careers

These modules provided an overview of how science communication strategies can be applied to science advocacy and policy, building relationships with policymakers, using power mapping for crafting effective messages, and how science policy awareness and analysis training can be applied to careers in non-academic and industry sectors.

Module 8: Leveraging Science Policy & Advocacy for Non-academic and Industry Careers. Speakers: Matthew Hanson (UCI), Chelsea Gregg (Medtronic), Stephanie Davis (NHLBI, JSPG), Kimberly Maxfield (FDA), Alan Arnold (MSRDC), Justin Clifford Smith (Optum, JSPG). Moderator: Harinder Singh (UC Irvine).

Local Engagement & Advocacy in Action

These modules presented options for applying science policy and advocacy knowledge at the local level, improve skills in scientific literacy and public engagement, and pitch research to state legislators.

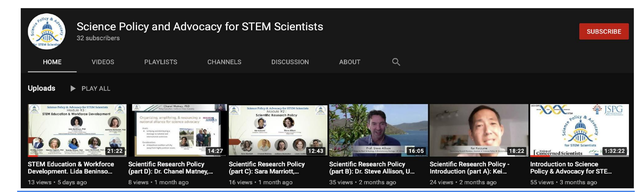
Module 9: Science Policy & Advocacy Engagement at the Local Level. Speakers: Karina Brocco French (UC Irvine), Chloe McPherson (AAME), Arti Garg (ESAL), Adam Fagen (ASTC), Geoffrey Hunt (ASM). Moderator: Rosie Dutt (Wash U St. Louis, JSPG).

Program Deliverables - Science Policy Podcast

Science Policy Podcast helps to connect policy & advocacy course participants with science policy professionals working at the frontiers of policymaking, Science & Technology policy fellows, UCI alumni in science policy circles and fellow students & postdocs actively engaged in the policy arena.

Program Deliverables - SciPol YouTube Channel

All the course lectures and panel discussion recordings are posted in a dedicated YouTube channel for science policy. The science policy community can access these publicly available lectures. In the future, lecture recordings will be used to teach a 2021 course in a "flipped-classroom learning" format.



Policy Elevator Pitch Competition

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